

Assignment 3

Muhammad Anas
23L-3005

R6. Yes, it is possible. Application layer can implement its own reliability mechanisms on top of UDP. Application layer can implement TCP like features like ACKs, sequence numbers, error detection, flow control, congestion control.

R15. a. First segment sequence no. = 90
second segment sequence no. = 110
 $110 - 90 = 20$
Data in first segment is 20 bytes

b. TCP uses cumulative acknowledgements. The receiver acknowledges the next byte it expects in order.

Host B expects byte 90 first but the segment starting with 110 arrives. But TCP only acknowledges in order data, so host B will still expect byte 90. So the acknowledgement no. sent by B will be 90.

P2 1) segment sent from server B $\rightarrow$ client C

Client C sent:

source port: 7532

dest port: 80

Source IP: C

dest IP: B

server's reply will be

source port: 80

dest port: 7532

Source IP: B

dest IP: C

2) Segment from Server B $\rightarrow$ Client C

Client opened another connection

source port = 26145

dest port = 80

source IP = C

dest IP = B

Server's reply:

source port = 80

dest port = 26145

source IP = B

dest IP = C

3) segment from server ~~B~~ $\rightarrow$ client A

Client A sent:

source port = 26145

dest port = 80

source IP = A

dest IP = B

Server's reply:

source port = 80

dest port = 26145

source IP = B

dest IP = A

P3.

01010011

01100110

01110100



01010011

01100110

10111001

+ 01110100

100101101



00101101

+

00101110

00101110

1's complement: 11010001
Checksum: 11010001

- Why UDP ~~uses~~ takes 1's complement and ^{receiver} error detection.

In the 1's complement method, sender sends checksum. The receiver ~~adds~~ data words and checksum. If no error occurred, the result will be all 1s. If we only sent the sum instead of 1's complement of sum, the receiver would have no simple way to verify correctness.

- 1 bit errors will always be detected because a single bit flip will change the sum, so the result will not equal all 1s.

- Yes, 2 bit error can go undetected. One bit flips from 0 to 1, Another from 1 to 0. Two changes can cancel each other, producing the same checksum.

P3/.
$$\text{Estimated RTT} = (1 - \alpha) \times \text{Estimated RTT} + \alpha \times \text{Sample RTT}$$

$$\text{DevRTT} = (1 - \beta) \times \text{DevRTT} + \beta \times \left| \text{sample RTT} - \text{Estimated RTT} \right|$$

$$\text{Timeout Interval} = \text{Estimated RTT} + 4 \times \text{DevRTT}$$

$$\alpha = 0.125, \quad \beta = 0.25$$

$$\text{Initial Estimated RTT} = 100 \text{ ms}$$

$$\text{Initial DevRTT} = 5 \text{ ms}$$

$$1) \text{ Sample RTT} = 106 \text{ ms}$$

$$\text{Estimated RTT} = 0.875(100) + 0.125(106)$$

$$= 100.75 \text{ ms}$$

$$\text{Dev RTT} = 0.75(5) + 0.25|106 - 100.75|$$

$$= 5.0625 \text{ ms}$$

$$\text{Timeout Interval} = 100.75 + 4(5.0625)$$

$$= 121 \text{ ms}$$

$$2) \text{ Sample RTT} = 120 \text{ ms}$$

$$\text{Estimated RTT} = 0.875(100.75) + 0.125(120)$$

$$= 103.156 \text{ ms}$$

$$\text{Dev RTT} = 0.75(5.0625) + 0.25|120 - 103.156|$$

$$= 8.0078 \text{ ms}$$

$$\text{Timeout Interval} = 103.156 + 4(8.0078)$$

$$= 135.19 \text{ ms}$$

$$3) \text{ Sample RTT} = 140 \text{ ms}$$

$$\text{Estimated RTT} = 0.875(103.156) + 0.125(140)$$

$$= 107.7615 \text{ ms}$$

$$\text{Dev RTT} = 0.75(8.0078) + 0.25|140 - 107.762|$$

$$= 164.02 \text{ ms}$$

$$4) \text{ Sample RTT} = 90 \text{ ms}$$

$$\text{Estimated RTT} = 0.875(107.762) + 0.125(90)$$

$$= 105.542 \text{ ms}$$

$$\text{Dev RTT} = 0.75(14.0654) + 0.25|90 - 105.54$$

$$= 14.4346 \text{ ms}$$

$$\text{Timeout Interval} = 105.542 + 4(14.4346)$$

$$= 163.28 \text{ ms}$$

5) Sample RTT = 115 ms
 Estimated RTT = $0.875(105.542) + 0.125(115)$
 $= 106.724 \text{ ms}$

DevRTT = $0.75(14.4346) + 0.25(115 - 106.724)$
 $= 12.8949 \text{ ms}$

Timeout Interval = $106.724 + 4(12.8949)$
 $= 158.3 \text{ ms}$

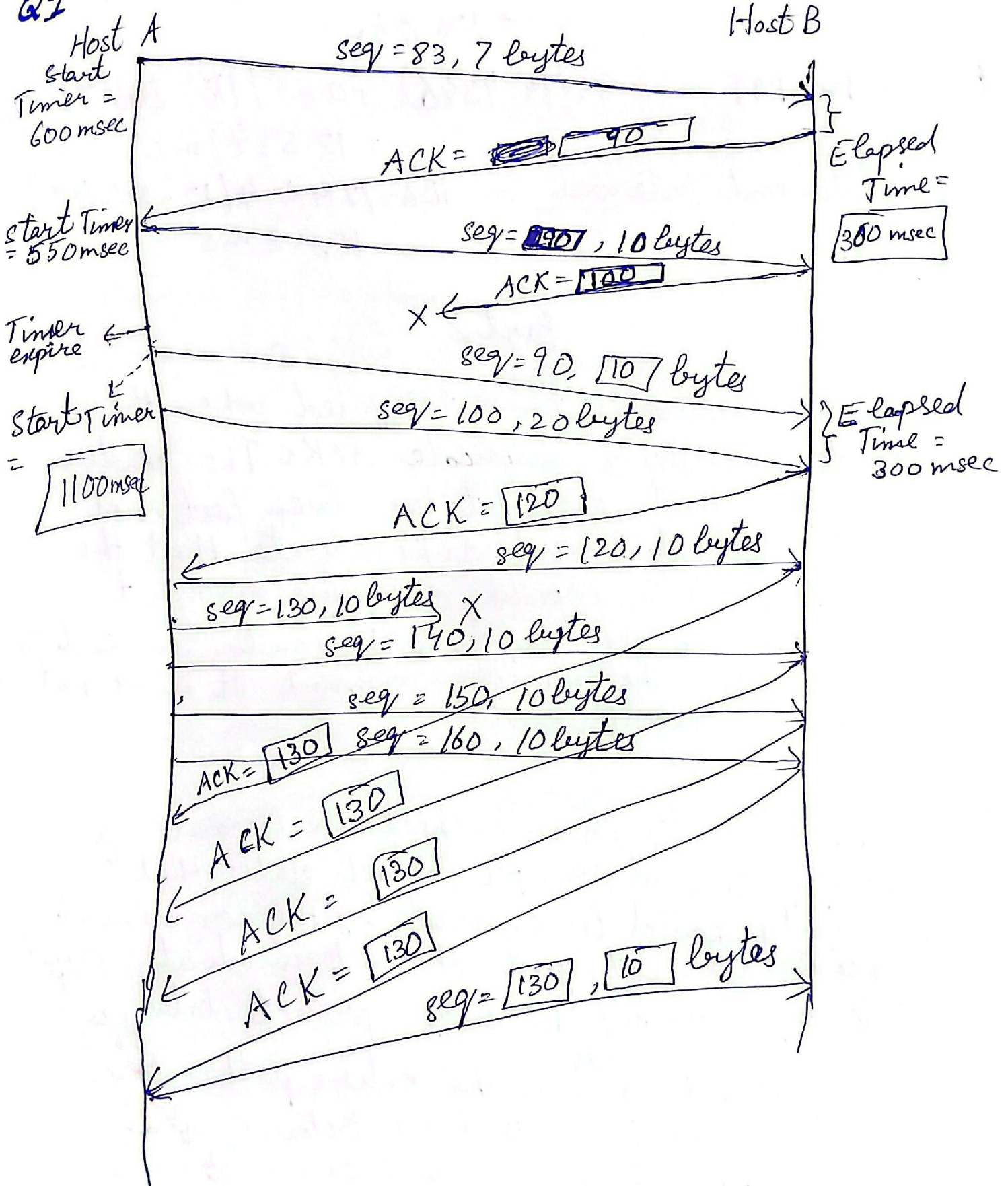
Part 2

Q2 Fast retransmit is triggered when the sender receives 3 duplicate ACKs. The sender assumes that a packet has been lost, not just delayed. Duplicate ACKs indicate that the receiver is still expecting a specific segment. In response to the trigger, the sender immediately retransmits the missing segment. It does not wait for the timeout.

Q3 UDP includes its own checksum because of the End-to-End principle, which states that reliability should be ensured by the communicating endpoints also, not just the intermediate layers. Link-layer checks, like CRC only protect data on a single link, not across the entire path. Errors can still occur in routers or between links. Therefore, UDP's checksum provides end-to-end error detection. It is not redundant.

Part 2

Q1



Q4 i) In selective repeat, only lost packets are transmitted.

Packets 5 and 9 are lost, only they will be retransmitted.

Total 2 retransmitted packets.

ii) In Go-Back N, once packet is lost, all subsequent packets are discarded by the receiver

$N=5$

Packet 6 is lost, so all subsequent packets within window N will be retransmitted.

Packets 6, 7, 8, 9, 10 will be retransmitted.

Total 5 packets

Q5